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September 25, 1978
SUPERSEDING
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December 16, 1970

FEDERAL SPECIFICATION

SPATULA, LABORATORY (Metal Blade)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers metal blade, laboratory spatulas.

1.2 Classification. The spatulas covered by this specification shall be of the following classes, grades and sizes, as specified (see 6.2).

Class I - Spatulas, all metal

Grade A - Rigid, nickel-silver blade

Size 1 - 3-1/2 inch blade

Grade B - Flexible, steel blade

Size 1 - 4 inch blade
Size 2 - 5 inch blade
Size 3 - 6 inch blade

Class II - Spatulas, flexible, steel blade, wood handle

Size 1 - 3 inch blade
Size 2 - 4 inch blade
Size 3 - 6 inch blade
Size 4 - 8 inch blade
Size 5 - 10 inch blade
Size 6 - 12 inch blade

FSC 6640

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Class III - Spatula, Steel, Semi-micro, Double Blade

Size 1 - 7 inch overall length

Class IV - Spatula, weighing, flexible, steel blade

Size 1 - 3 inch blade

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Military Specification

MIL-P-116 - Preservation Packaging, Methods Of.

Military Standards

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military specification, standards, required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

Federal Specifications

PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Boxes, Fiberboard.
PPP-B-676 - Boxes, Set-Up, Paperboard

Federal Standard

FED-STD-123 - Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

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(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston; New York; Atlanta; Chicago; Kansas City, Mo.; Fort Worth; Denver, San Francisco; Auburn, Washington; and Washington, D.C.).

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies).

2.2 Other Publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bid or request for proposal shall apply.

AISI Requirements for Stainless and Heat-Resisting Steel.

(Application for copies should be addressed to the American Iron and Steel Institute, 150 E. 42nd Street, New York, New York 10017).

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM B 122	-	Copper-Nickel-Zinc Alloy (Nickel Silver) and Copper-Nickel Alloy Plate, Sheet, Strip, and Rolled Bar, Spec. for.
ASTM E 18	-	Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials, Tests for.
ASTM E 38	-	Nickel-Chromium and Nickel-Chromium-Iron Alloys, Chemical Analysis of.
ASTM E 75	-	Copper-Nickel and Copper-Nickel-Zinc Alloys, Chemical Analysis of.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

3. REQUIREMENTS

3.1 Material.

3.1.1 Nickel silver. The chemical composition of nickel silver shall be in accordance with ASTM Designation B 122 for copper alloy No. 770, nickel silver, 18% alloy B. The composition analysis of nickel silver shall be determined as specified in 4.4.1. The nickel-silver shall show no evidence of corrosion when tested as specified in 4.4.2.1.

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3.1.2 Aluminum. The aluminum shall be a cast aluminum.

3.1.3 Stainless steel.

3.1.3.1 Martensitic. The martensitic steel shall conform to the requirements of AISI 420 or 440 for wrought stainless steel. The composition analysis shall be determined as specified in 4.4.1.2. The martensitic steel shall show no evidence of corrosion when tested as specified in 4.4.2.1.

3.1.3.2 Precipitation-hardening. The precipitation-hardening steel shall conform to the requirements of AISI 631, wrought stainless steel. The composition hardening steel shall show no evidence of corrosion when tested as specified in 4.4.2.1.

3.1.4 Wood. The wood shall be a durable, smoothly finished hardwood. The wood used shall be one of the following species: beechwood, rosewood, cocobolo, or walnut.

3.2 Construction and design.

3.2.1 Class I, Grade A. The blade shall be fabricated of nickel silver as specified in 3.1.1. The blade shall fit into the handle and shall be securely attached to it. The width of the blade at the handle shall be not less than 0.56 inch (in.). The end of the blade opposite the handle shall be rounded and smoothly finished.

3.2.1.2 Handle. The handle shall be fabricated of cast aluminum as specified in 3.1.2 or stainless steel as specified in 3.1.3.1 or 3.1.3.2. The handle shall be a nominal 0.50 in. in thickness and 0.69 inch in width, tapering to 0.56 in. in width at the juncture of the handle and the blade. The sides and end of the handle shall be rounded and smoothly finished.

3.2.2 Class I, Grade B.

3.2.2.1 Blade. The blade shall be fabricated of stainless steel as specified in 3.1.3.1 or 3.1.3.2. The blade shall be flexible and shall not break or take a permanent set when tested as specified in 4.4.2. The blade shall fit into the handle and shall be securely attached to it. At least one-half the length of the blade shall have a uniform taper in thickness. The end of the blade shall be rounded and smoothly finished.

3.2.2.2 Handle. The handle shall conform to the requirements specified in 3.2.1.2.

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3.2.3 Class II and Class IV.

3.2.3.1 Blade and tang. The blade and tang shall be fabricated of stainless steel as specified in 3.1.3.1 or 3.1.3.2. The tang (that part of the blade that fits inside of the handle) shall extend at least one-half the length of the handle and shall be the same width as the handle. The sides of the tang shall be smoothly ground to conform to the contour of the handle. At least one-half of the length of the blade shall have a uniform taper in thickness. The blade shall be flexible and shall not break, show evidence of fatigue, or take a permanent set from its original position when tested as specified in 4.4.3. The blade shall have a Rockwell hardness of C40 to C50 when tested as specified in 4.4.4. The tip of the blade shall be rounded. The class IV blade shall be tapered and rounded at the tip.

3.2.3.2 Handle. The handle shall be fabricated of hardwood as specified in 3.1.4. The handle shall be slotted to provide for the attachement of the tang. The blade and tang shall be securely attached to the handle by two telescoping type rivets. The rivets shall be fabricated of corrosion-resisting steel or brass.

3.2.4 Class III. Class III spatulas shall be fabricated from stainless steel wire conforming to the requirements in 3.1.3.1 or 3.1.3.2. The ends of the wire shall be flattened in blades. One blade shall have a rounded end and the other blade shall have a square end. The wire between the two blades shall serve as the handle.

3.3 Dimensions. The dimensions of the spatulas shall be as specified in table I for the appropriate classes, grades and sizes. Dimensions shall be expressed in inches.

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TABLE I. Dimensions of spatulas.

Class	Grade	Size	Length of blade	Width of blade	Overall length	Minimum thickness of blade*
I	A	1	3-1/2 ± 1/8	0.94	7-1/2 ± 1/4	0.040
I	B	1	4 ± 1/8	0.75	7 ± 1/4	0.049
I	B	2	5 ± 1/8	0.87	8 ± 1/4	0.058
I	B	3	6 ± 1/4	1.00	9-1/2 ± 1/4	0.058
II	--	1	3 ± 1/8	0.62	6-3/8 ± 1/4	0.049
II	--	2	4 ± 1/8	0.75	7-3/8 ± 1/4	0.049
II	--	3	6 ± 1/8	1.00	10 ± 1/4	0.055
II		4	8 ± 1/4	1.25	12-1/2 ± 1/4	0.065
II	--	5	10 ± 1/4	1.50	14-3/4 ± 1/4	0.065
II	--	6	12 ± 1/4	1.75	17-1/2 ± 1/4	0.083
III	--	1	2 ± 1/8	5/16 ± 1/16	7 ± 1/4	-----
IV	--	1	3 ± 1/8	0.56	6-3/8 ± 1/4	0.049

* The blade thickness shall be measured at thickest cross section of the tang.

3.3.1 Tolerances. Unless otherwise specified in table I, the tolerance for the width of the blade shall be plus 0.13, minus 0.07 in.

3.4 Finish.

3.4.1 Blade. The blade of all spatulas shall be ground to remove all surface imperfections and polished to a No. 8 mirror finish. The No. 8 mirror finish shall be as defined in AISI requirements for Stainless and Heat Resisting Steels.

3.4.2 Handle. The hardwood handle shall be smoothly finished and polished.

3.5 Identification marking. The blade of each spatula shall be legibly and permanently marked with the name or registered trademark of the manufacturer.

3.6 Workmanship. Workmanship shall be first class throughout. The spatulas shall be free from defects which detract from their appearance or impair their serviceability.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technial means of physical and chemical properties) of the item.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and table II. The unit of product for sampling purposes shall be one spatula.

TABLE II. Sampling for examination.

	Inspection level	AQL (percent defective
For visual examination		
Major defects	II	1.0
Minor defects	II	2.5
For dimensional examination	S-3	4.0

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and table III. The inspection level shall be S-2 and the acceptance number shall be zero. Unit of product for sampling purposes shall be one spatula and for chemical analysis shall be four ounces of each lot of each metal used in the construction of the spatulas.

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TABLE III. Sampling for tests.

Characteristic	Requirement	Test procedure
Chemical analysis		
Nickel silver	3.1.1	4.4.1.1
Martensitic	3.1.3.1	4.4.1.2
Precipitation-Hardening	3.1.3.2	4.4.1.3
Resistance to corrosion		
Nickel silver	3.1.1	4.4.2.1
Martensitic	3.1.3.1	4.4.2.1
Precipitation-Hardening	3.1.3.2	4.4.2.1
Flexibility	3.2.3.1	4.4.3
Rockwell hardness	3.2.3.1	4.4.4

4.2.2.1 Where possible, the same sample unit shall be used for the determination of two or more test characteristics.

4.3 Examination. The spatulas shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the following classification of defects listed in table IV.

TABLE IV. Classification of defects.

Categories	Defects*
Major	
101	Spatula not free from sharp edges.
102	Spatula not free from deformation.
103	Blade not securely attacned to handle.
104	Metal handle and blade not free from pits, burrs, chips, or scratches.
105	Wood handle not smooth or free from scratches, dents, or rough edges.
106	Wood handle not free from cracks, chips, or blisters.
Minor	
201	Flexible, steel-blade spatulas not free from grind marks.
202	Identification marking not present, complete, correct, legible, or permanent.
203	Blade and handle not free from dirt, grease, or foreign particles.

* Examination shall not be restricted to the classified possible defects listed above.

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4.3.2 Dimensional examination. The spatulas shall be examined for defects in dimensions. Any dimension not within the tolerance specified herein shall be classified as a defect.

4.3.3 Examination of preparation for delivery. An examination shall be made to determine compliance with the packaging, packing and marking requirements of section 5. Defects shall be scored as specified in table V. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 with AQL of 4.0 expressed in terms of percent defective.

TABLE V. Examinaton of preparation for delivery.

Examine	Defect
Contents	Not as specified.
Container	Not as specified.
Marking	Omitted; incorrect; illegible; improper size, location, sequence or method of application.
Material	Component missing or damaged.
Workmanship	Bulging or discoloration of containers.

4.3.3.1 Examination of closure, waterproofing and banding of containers. When the containers are required to comply with PPP-B-566, PPP-B-636, or PPP-B-676 the examination for defects in closure, waterproofing or banding shall be in accordance with the appendix of the box specification.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Chemical analysis.

4.4.1.1 Nickel silver. The chemical composition of nickel silver shall be determined in accordance with ASTM Designation E-75.

4.4.1.2 Martensitic stainless steel. The chemical composition of the martensitic stainless steel shall be determined in accordance with ASTM Designation E-38.

4.4.1.3 Precipitation-Hardening stainless steel. The chemical composition of AISI 631 stainless steel shall be determined in accordance with ASTM Designation E-38.

4.4.2 Resistance to corrosion.

4.4.2.1 Boil test. Samples shall be scrubbed with soap and water, rinsed thoroughly in distilled water, and dried. After cleaning, samples shall be boiled in distilled water for 30 minutes and allowed to remain submerged for an additional 24 hours.

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4.4.3 Flexibility test. Except for class III, all spatulas with steel blades shall be subjected to a bending test as follows: The blade shall be secured in a vise halfway between the tip and the handle. With the handle down, the working end shall be bent six times in each direction from the vertical to 30° for size 1, 2 and 3, and 45° for sizes 4, 5 and 6.

4.4.4 Rockwell hardness. The Rockwell hardness of the blade shall be determined in accordance with ASTM Designation E-18.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, or C (Commercial), as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit package. Each spatula shall be clean, dry and packaged in accordance with MIL-P-116, method Ic-1 or method IV, except that a preservative shall not be used.

5.1.1.2 Intermediate package. Twelve unit packages containing spatulas for one class, grade and size only shall be packaged in a box conforming to PPP-B-566 or PPP-B-676. Closure shall be as specified in the appendix of the box specification.

5.1.2 Level C (Commercial). Each spatula shall be clean, dry and packaged in a manner that will afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B or C (commercial) as specified (see 6.2).

5.2.1 Level A. Twenty-four intermediate packages containing spatulas of one class, grade and size only of class I, grade A, size 1; class I, grade B, sizes 1, 2 or 3; class II, sizes 2 or 3; class III, size 1; class IV, size 1; or twelve intermediate packages containing one size only of class II, sizes 4, 5 or 6 or forty-eight intermediate containers containing class II, size 1 spatulas shall be packed in a shipping container designed for a type 2 load and conforming to PPP-B-636, class weather-resistant. Closure, bonding and waterproofing shall be as specified in the appendix of the box specification.

5.2.2 Level B. Spatulas shall be packed as specified in 5.2.1 except that the shipping container shall conform to PPP-B-636, class domestic and bonding and waterproofing shall not be required.

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5.2.3 Level C (commercial). The packaged spatulas shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the carriers' rules and regulations.

5.3 Marking.

5.3.1 Civil agencies. Marking shall be as specified in the contract or order.

5.3.2 Military activities. Each unit package, intermediate and shipping container shall be marked as specified in MIL-STD-129.

6. NOTES

6.1 Intended use. The spatulas covered by this specification are intended for various pharmaceutical, chemical, and dental applications.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Class, grade, and size required (see 1.2).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 This specification does not include all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet requirements.

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6.4 This specification covers the following items shown in the Federal Supply Catalog:

National Stock Number	Item Identification
6640-00-171-5198	SPATULA, LABORATORY, 4 inch Blade.
6640-00-264-5032	SPATULA, LABORATORY, 1-3/4 inch Blade.
6640-00-264-8285	SPATULA, LABORATORY, 12 inch Blade.
6640-00-264-8286	SPATULA, LABORATORY, 10 inch Blade.
6640-00-439-7350	SPATULA, LABORATORY, 3 inch Blade.
6640-00-439-7365	SPATULA, LABORATORY, 6 inch Blade.
6640-00-439-7375	SPATULA, LABORATORY, 8 inch Blade.
6640-00-556-9000	SPATULA, LABORATORY, 3-1/2 inch Blade.
6640-00-975-0453	SPATULA, LABORATORY, 3 inch Blade.

Military custodians:
Army - MD
Navy - MS
Air Force - 03

Review activity:
DoD - MB

Preparing activity:
Defense Logisitcs Agency - DM

Civil Agency Coordinating Activities
GSA - FSS
VA - DMS
HEW - NIH

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